

Work Stream 5: Incompressibility vs. Divergence-Free Vectors

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1 Introduction

A common point of confusion in interdisciplinary fluid dynamics, raised insightfully by u/brokebeany, is the distinction between a fluid being *divergence-free* and a fluid being *incompressible*.

2 The Mathematical vs. Physical Gap

In the OpenAI Lean 4 proof, the constructed vector field $u(x, t)$ satisfies:

$$\nabla \cdot u = 0 \quad \text{for all } t \in [0, 1) \tag{1}$$

Mathematically, this means the PDE is perfectly satisfied. However, in physics, the equation $\nabla \cdot u = 0$ is derived from the continuity equation $\partial_t \rho + \nabla \cdot (\rho u) = 0$ by assuming that density ρ is strictly constant.

This assumption is an asymptotic limit that only holds when the Mach number $Ma = |u|/c_s \ll 1$. When the fluid in the OpenAI singularity breaches $Ma = 0.3$ (at roughly $\tau = 67$ femtoseconds), physical density variations scale as $\Delta\rho/\rho \sim Ma^2 \sim 0.1$.

Therefore, while the *vector field* remains mathematically divergence-free, the *physical fluid* it is supposed to represent becomes compressible. The equation stops matching reality.

3 Acknowledgments

We thank u/brokebeany for prompting this critical clarification on the ontology of fluid mechanics.